SIMATS ENGINEERING
ASSESSMENT TOOL 2- COMPARATIVE ANALYSIS

Course Code:	**CSA12**	**Course Title:**	**Computer Architecture**
CO Assessed:	**CO4**	**Assessment:**	**ASSESSMENT TOOL2- COMPARATIVE ANALYSIS**
Weightage:	**30%**	**Total Marks:**	**25**
CO4 Statement:	*Implement hierarchical memory systems by differentiating cache levels (L1, L2, L3), virtual memory with paging, and advanced memory technologies (DRAM, SRAM, NAND Flash, MRAM, RRAM) based on latency, bandwidth, and throughput characteristics.*		
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ANNEXURE A
COMPARATIVE ANALYSIS

1. Compare L1, L2, and L3 cache levels in terms of latency, bandwidth, and throughput, and analyze their impact on processor performance.

Comparison of L1, L2, and L3 Cache

Cache memory is a small, high-speed memory located close to or inside the processor. It stores frequently accessed instructions and data so that the CPU does not always need to access slower main memory.

Feature	**L1 Cache**	**L2 Cache**	**L3 Cache**
Location	Closest to CPU core	Close to/inside CPU core	Usually shared among cores
Typical Size	Few KB to ~128 KB per core	Hundreds of KB to a few MB	Several MB to tens of MB
Latency	Lowest, about 1–4 CPU cycles	Higher, about 4–15 cycles	Higher, about 20–50+ cycles
Bandwidth	Very high	High	Lower than L1/L2
Throughput	Highest	Very high	High
Cost per bit	Highest	High	Lower
Purpose	Immediate CPU data/instructions	Secondary fast storage	Large shared cache
Access frequency	Very high	High	Moderate

L1 Cache

L1 is the fastest cache and is normally divided into instruction cache (L1-I) and data cache (L1-D).

Because it is very small, it can be accessed extremely quickly. A successful L1 access, called an **L1 hit**, allows the processor to continue with minimal delay.

L2 Cache

L2 is larger than L1 but slower. If the required data is not found in L1, the processor checks L2. L2 provides a balance between capacity and speed.

L3 Cache

L3 is larger and slower than L1 and L2. In modern multicore processors, L3 is commonly shared among several or all CPU cores. It reduces the number of expensive accesses to DRAM.

Impact on Processor Performance

The effectiveness of cache is strongly related to the **cache hit rate**.

- A high L1 hit rate gives extremely low access latency.
- L2 catches many accesses that miss in L1.
- L3 reduces accesses to relatively slow main memory.
- Better cache locality increases processor throughput.
- A cache miss causes additional latency because data must be fetched from the next memory level.

Therefore, a processor with effective cache organization can execute instructions much faster than one that frequently accesses DRAM.

Conclusion: L1 provides the lowest latency and highest bandwidth, L2 provides greater capacity with slightly higher latency, and L3 provides large shared capacity while still being much faster than main memory.

2. Differentiate SRAM and DRAM by evaluating their latency, bandwidth, and throughput characteristics in cache and main memory design.

SRAM vs DRAM

SRAM (Static Random Access Memory) and **DRAM (Dynamic Random Access Memory)** are both volatile semiconductor memories, but they have different structures and performance characteristics.

Feature	**SRAM**	**DRAM**
Storage element	Flip-flop, typically 6 transistors	Capacitor + transistor
Latency	Very low	Higher
Bandwidth	Very high	High
Throughput	Very high for small, frequent accesses	High for large sequential transfers
Refresh required	No	Yes
Density	Low	High
Cost per bit	High	Low
Power per bit	Relatively high area/cost	Lower area per bit, but refresh consumes energy
Typical use	CPU caches	Main memory

SRAM in Cache Design

SRAM is normally used for L1, L2, and L3 caches because it provides very fast access. Its stored data remains available as long as power is supplied, without periodic refresh.

The main advantage is **low latency**. This is important because CPUs operate at extremely high clock frequencies and cannot afford frequent long waits for memory.

However, SRAM requires more transistors per bit, making it relatively large and expensive. Therefore, building a very large SRAM cache would significantly increase processor size and cost.

DRAM in Main Memory

DRAM stores each bit using a transistor and capacitor. The capacitor gradually loses its charge, so DRAM must periodically be refreshed.

DRAM has higher latency than SRAM, but it offers much greater density at a lower cost per bit. This makes it suitable for large-capacity main memory.

Modern DRAM systems also use wide memory buses, multiple channels, and burst transfers to achieve high bandwidth.

Latency, Bandwidth and Throughput

Latency: SRAM is significantly faster because its storage cells can be accessed without the sensing and refreshing operations associated with DRAM.

Bandwidth: SRAM can provide extremely high bandwidth when placed close to the processor. DRAM can also provide high aggregate bandwidth through multiple channels and high-speed interfaces, but individual access latency remains higher.

Throughput: SRAM is excellent for small, random, repeated accesses typical of CPU caches. DRAM achieves good throughput when transferring large blocks of data.

Conclusion

SRAM is preferred for **cache memory** because speed is more important than capacity. DRAM is preferred for **main memory** because capacity and cost efficiency are more important.

The basic hierarchy is:

CPU → SRAM Cache → DRAM Main Memory → Secondary Storage

This combination provides a good balance between speed, capacity, and cost.

3. Compare virtual memory with paging and cache memory mechanisms in terms of access latency and system throughput.

4. Analyze the differences between NAND Flash and DRAM with respect to latency, bandwidth, and suitability for hierarchical memory systems.

Virtual Memory, Paging, and Cache Memory

Virtual memory and cache memory both improve the usability and performance of computer systems, but they solve different problems.

Feature	Virtual Memory	Paging	Cache Memory
Main purpose	Gives processes a large virtual address space	Manages virtual-to-physical memory mapping	Reduces CPU memory access time
Storage involved	RAM + secondary storage	RAM + secondary storage	SRAM cache + RAM
Typical latency	High when disk/SSD access occurs	High on a page fault	Very low on a cache hit

Feature	Virtual Memory	Paging	Cache Memory
Unit of transfer	Pages	Fixed-size pages	Cache blocks/lines
Main benefit	Memory abstraction and isolation	Efficient memory allocation	Faster data access
Performance issue	Page faults	Page faults and translation overhead	Cache misses

Virtual Memory

Virtual memory allows a process to use a virtual address space that can be larger than the available physical RAM. The operating system maps virtual addresses to physical memory.

When required data is already in RAM, access can proceed normally after address translation. However, if the required page is not in RAM, a **page fault** occurs.

The operating system must retrieve the page from secondary storage, which can introduce extremely large delays compared with RAM access.

Paging

Paging divides virtual and physical memory into fixed-size units called **pages** and **page frames**.

A page table records the mapping between virtual pages and physical frames. A **Translation Lookaside Buffer (TLB)** is used to cache recent address translations.

A TLB hit can make address translation fast. A TLB miss requires additional page-table accesses, increasing latency.

If the page itself is absent from physical memory, a page fault occurs. This is much more expensive than either a TLB miss or a cache miss.

Cache Memory

Cache memory stores recently or frequently used data close to the processor.

When the CPU requests data:

1. The processor checks L1 cache.
2. If it misses, it checks L2.
3. If L2 misses, L3 may be checked.
4. Eventually, main memory may be accessed.

A cache hit has very low latency, while a cache miss increases access time.

Effect on Throughput

Cache memory primarily improves **CPU throughput** by reducing the average memory access time.

Virtual memory primarily provides **capacity, protection, isolation, and address-space flexibility**, rather than making memory faster.

Paging improves memory utilization by allowing only the currently needed pages to occupy physical memory. However, excessive page faults can cause **thrashing**, where the system spends much of its time moving pages instead of executing programs.

Overall Relationship

The mechanisms work together:

CPU → Cache → TLB → RAM → Virtual Memory/Storage

Cache reduces the time needed to access frequently used data, while paging and virtual memory manage how programs use physical memory.

Conclusion: Cache memory is primarily a **performance mechanism**, whereas virtual memory and paging are primarily **memory-management mechanisms**. Cache hits can take only a few CPU cycles, while page faults can be many orders of magnitude slower.

5. Compare MRAM and RRAM technologies by examining their performance in terms of access time, throughput, and energy efficiency.

MRAM vs RRAM

MRAM (Magnetoresistive Random Access Memory) and **RRAM/ReRAM (Resistive Random Access Memory)** are emerging non-volatile memory technologies. Both aim to provide fast, persistent memory while reducing some limitations of conventional memory technologies.

Feature	**MRAM**	**RRAM**
Storage principle	Magnetic state	Resistance state
Volatility	Non-volatile	Non-volatile
Access time	Very fast	Very fast to moderate, depending on implementation
Read performance	High	High
Write performance	High	Potentially high
Throughput	High	High potential
Energy efficiency	Good	Potentially very good
Endurance	Very high	Generally high, but device-dependent
Data retention	Excellent	Good to excellent, device-dependent
Density	Good	Potentially very high
Technology maturity	Relatively mature	More emerging
Potential applications	Embedded memory, cache, persistent memory	High-density memory, neuromorphic and storage-class applications

MRAM

MRAM stores information using magnetic states rather than electrical charge.

A common form is **STT-MRAM (Spin-Transfer Torque MRAM)**, while newer approaches include **SOT-MRAM (Spin-Orbit Torque MRAM)**.

MRAM has several important advantages:

- Non-volatility
- Fast read access
- Very high endurance
- Good energy efficiency
- No refresh operation
- Good resistance to data loss when power is removed

Because MRAM does not require periodic refresh like DRAM, it can reduce standby energy consumption.

MRAM can potentially be used in embedded memory and, depending on the implementation and system requirements, as an alternative to some SRAM or DRAM applications.

RRAM

RRAM stores information by changing the electrical resistance of a material. A high-resistance state and low-resistance state can represent different data values.

RRAM is attractive because memory cells can potentially be very small, giving it high-density potential.

Important advantages include:

- Non-volatility
- Simple cell structures
- Potentially high density
- Low-energy operation in suitable implementations
- Potential for 3D integration
- Potential for analog and neuromorphic computing

However, RRAM technology can face challenges involving device variability, resistance-state stability, endurance, and manufacturing consistency.

Access Time

MRAM can provide very fast read access, making it attractive for applications requiring persistent data with relatively low latency.

RRAM can also provide fast access, but actual performance varies considerably with the cell technology, read/write circuitry, array architecture, and operating conditions.

Therefore, neither technology should be judged only by the memory cell itself. **System-level access time** also depends on controllers, interconnects, sensing circuits, and memory architecture.

Throughput

Both technologies can achieve high throughput when implemented with parallel memory arrays.

MRAM benefits from fast read/write operation and high endurance, while RRAM's high-density arrays provide strong potential for high parallel throughput.

In practice, throughput depends on:

- Memory-array organization
- Number of banks
- Bus width
- Parallelism
- Controller architecture
- Read/write timing
- Technology generation

Energy Efficiency

MRAM has an important energy advantage because it is non-volatile and does not require refresh.

RRAM also has significant potential for low-energy operation because resistance switching can require relatively little energy, although actual energy consumption depends strongly on the switching mechanism and device design.

MRAM vs RRAM Summary

MRAM is particularly attractive when:

- Very high endurance is required.
- Fast access is important.
- Non-volatility is needed.
- Reliable embedded memory is required.

RRAM is particularly attractive when:

- Very high density is required.

- Low power is important.
- 3D integration is desirable.
- New computing architectures such as neuromorphic systems are being considered.

Technology	Relative Latency	Bandwidth/Throughput	Volatile?	Main Application
L1 Cache / SRAM	Extremely low	Extremely high	Yes	CPU immediate-access cache
L2 Cache / SRAM	Very low	Very high	Yes	Secondary CPU cache
L3 Cache / SRAM	Low compared with DRAM	High	Yes	Large shared CPU cache
DRAM	Low–moderate	Very high	Yes	Main memory
NAND Flash	High	High, especially sequential	No	SSD/storage
MRAM	Very low–low	High	No	Embedded/persistent memory
RRAM	Low–moderate, device-dependent	High potential	No	High-density/emerging memory

Code of Conduct:

I GOKUL RAJ 192521494_ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Scope of Items	20%	Selects highly relevant items with clear scope and justification	Selects appropriate items with minor gaps	Selection is limited or partially relevant	Items are unclear or not relevant
Comparison Depth	25%	Provides detailed and comprehensive comparison across multiple aspects	Provides adequate comparison with some depth	Limited comparison with few aspects	Very minimal or no comparison
Use of Evidence	20%	Supports comparison with accurate data, examples, or references	Uses some supporting evidence with minor gaps	Limited or weak supporting evidence	No supporting evidence provided
Critical Thinking	20%	Demonstrates strong critical thinking with clear interpretation and reasoning	Shows reasonable interpretation with minor gaps	Basic interpretation; lacks depth	No meaningful interpretation
Clarity of Presentation	15%	Well-organized, clear, and logically structured presentation	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand